Deep Reinforcement Learning

Deep Q Learning

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Content

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- Recap:
  - Tabular SARSA & Q-learning
  - Value function approximation with NNs
- On-policy control with gradients and semi-gradients
- Deep Q networks (DQN)
  - Extensions & modifications
- Least squares policy iteration

Where are we?

Chapter	Topic	Content
	Basics & tabular methods	
1-5	Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning	RL basics in finite dimensions
	Deep-learning-based methods	
6	Brief introduction to deep learning	The basics for what comes next
7	Value function approximation	Value estimation with function approximation
8	Deep Q-learning	Q-learning with neural networks
9	Policy gradients	
10	Actor-critic algorithms	
11	Advanced algorithms (Part I): From policy gradient to PPO	
12	Advanced algorithms (Part II): From Q-learning to Soft Actor-Critic	
	Model-Based Control	
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Lecture contents

Recap: Tabular SARSA and Q-learning

Tabular SARSA and Q-learning

Algorithm: SARSA (On-policy).
initialize

- $Q(s, a)$ arbitrarily for $s \in \mathcal{S}, a \in \mathcal{A}$
- $Q(\text{terminal}, \cdot) = 0$
- $\pi = \epsilon$-greedy(Q)

for $k = 1, 2, \ldots, K$ episodes:
  Initialize $s_t \leftarrow s_0, t \leftarrow 0$
  while s_t is not terminal:
    Take action $a_t \sim \pi(s_t)$ and observe (r_t, s_{t+1})
    Select $a_{t+1} \sim \pi(s_{t+1})$
    Update Q given $(s_t, a_t, r_t, s_{t+1}, a_{t+1})$:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)\right]$$

    Update policy: $\pi = \epsilon$-greedy(Q)
    $t \leftarrow t + 1$

Algorithm: Q-learning (Off-policy).
initialize

- $Q(s, a)$ arbitrarily for $s \in \mathcal{S}, a \in \mathcal{A}$
- $Q(\text{terminal}, \cdot) = 0$
- $\pi = \epsilon$-greedy(Q)

for $k = 1, 2, \ldots, K$ episodes:
  Initialize $s_t \leftarrow s_0, t \leftarrow 0$
  while s_t is not terminal:
    Take action $a_t \sim \pi(s_t)$ and observe (r_t, s_{t+1})
    ~~Select $a_{t+1} \sim \pi(s_{t+1})$~~
    Update Q given (s_t, a_t, r_t, s_{t+1}):

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)\right]$$

    Update policy $\pi = \epsilon$-greedy(Q)
    $t \leftarrow t + 1$

Tabular SARSA and Q-learning: Gridworld

On-policy: SARSA

Off-policy: Q-learning

- Discount: $\gamma = 0.9$
- Exploration: $\epsilon = 0.2$
- Step size: $\alpha = 0.5$

Recap: Value function approximation

- We want to **approximate the value function by parametric model** with trainable parameters $\theta \in \mathbb{R}^d$, i.e., $V_\theta(s) \approx V^\pi(s)$.
- For training, we need to define a **prediction objective**:

$$L(\theta) = \sum_{k=1}^{N} \Big(V^\pi(s_k) - V_\theta(s_k)\Big)^2 \approx \int_{\mathcal{S}} \mu(s) \Big(V^\pi(s) - V_\theta(s)\Big)^2 \,\mathrm{d}s = \overline{VE}(\theta). \tag{1}$$

- **Incremental weight updates** via
  - gradients, e.g., in the Monte Carlo setting:

$$\theta \leftarrow \theta + \alpha[g - V_\theta(s)]\nabla_\theta V_\theta(s). \tag{2}$$

  - semi-gradients in the bootstrapping case: we are not differentiating the target, which also depends on V_θ. For TD learning, we have

$$\theta \leftarrow \theta + \alpha[r + \gamma V_\theta(s') - V_\theta(s)]\nabla_\theta V_\theta(s). \tag{3}$$

- **Batch updates** using $\mathcal{B} = \{(s_i, r_i, s_i')\}_{i=1}^b \subset \mathcal{D}$ instead of a single sample as in SGD. For TD updates, we thus get

$$\theta \leftarrow \theta + \frac{\alpha}{b} \sum_{(s_i, r_i, s_i') \in \mathcal{B}} [r_i + \gamma V_\theta(s_i') - V_\theta(s_i)]\nabla_\theta V_\theta(s_i). \tag{4}$$

- For **linear models** (i.e., $V_\theta(s) = \theta^\top \psi(s) = \sum_{k=1}^d \theta_k \psi_k(s)$), training can be done in one step via linear regression:

$$\theta^* = (\Psi^\top \Psi)^{-1} \Psi^\top y = \Psi^\dagger y \qquad \Rightarrow \qquad V_{\theta^*}(s) = \theta^{*\top} \psi(s) = \sum_{k=1}^{d} \theta_k^* \psi_k(s). \tag{5}$$

On-policy control with gradients and semi-gradients

Value-based control: Q-function approximation

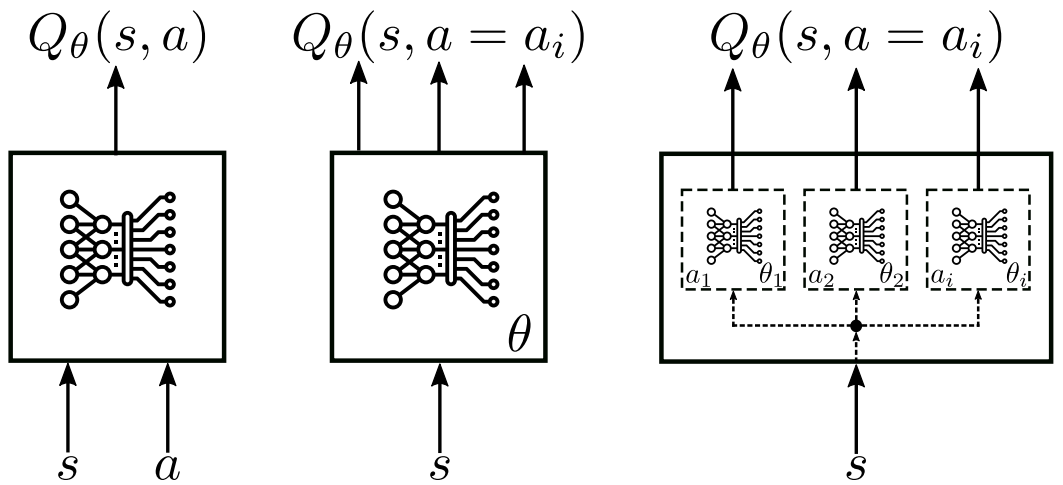
Today's focus: **value-based control** ⇒ How to transfer tabular methods to the function approximation case.

- The states are continuous, but the actions are (usually) still discrete.
- Central object of interest: The Q function that we wish to approximate with a parametric model (e.g., a neural net)

$$Q_\theta(s,a) \approx Q^\pi(s,a)$$

- We can then use this in the well-known generalized policy iteration (GPI) scheme to identify optimal policies:

$$Q_\theta(s,a) \approx Q^*(s,a) \quad \Rightarrow \quad \pi^*(s) = \arg\max_{a \in \mathcal{A}} Q_\theta(s,a).$$



Types of architectures (Abdelwanis et al. 2026)

Gradient-based action value learning

The transfer from value function approximation is straightforward!

- Adaptation of (1) to get $L(\theta)$

$$L(\theta) = \sum_{k=1}^{N} \Big(Q^{\pi}(s_k, a_k) - Q_{\theta}(s_k, a_k) \Big)^2. \tag{6}$$

- Incremental update using (semi-)gradients:

$$\theta \leftarrow \theta + \alpha [Q^{\pi}(s, a) - Q_{\theta}(s, a)] \nabla_{\theta} Q_{\theta}(s, a). \tag{7}$$

- Depending on the control approach, the true target $Q^{\pi}(s, a)$ is approximated by:
  - Monte Carlo: full episodic return

$$Q^{\pi}(s, a) \approx g,$$

  - SARSA: one-step bootstrapped estimate

$$Q^{\pi}(s, a) \approx r + \gamma Q_{\theta}(s', a'),$$

Algorithm: Gradient MC control

Algorithm: Gradient Monte Carlo algorithm for estimating Q^π / Q^*
Input:

- a differentiable, parameter-dependent function $Q_\theta : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$
- learning rate α
- **Prediction case**: a policy π
- **Improvement case**: ϵ defining the ϵ-greedy policy update based on Q_θ

Initialize: Value function weights $\theta \in \mathbb{R}^d$ arbitrarily

for $k = 1, 2, \dots, K$ episodes:
Generate a sequence following π (or ϵ-greedy(Q_θ)):

$$((s_0, a_0, r_0), (s_1, a_1, r_1), \dots, (s_{T_k-1}, a_{T_k-1}, r_{T_k-1}))$$

Calculate the every-visit returns g_t
for $t = 0, 1, \dots, T-1$:
$\theta \leftarrow \theta + \alpha[g_t - Q_\theta(s_t, a_t)]\nabla_\theta Q_\theta(s_t, a_t)$

- Direct transfer from tabular case to function approximation.
- Update target becomes the sampled return $Q^\pi(s_t, a_t) \approx g_t$.
- If operating ϵ-greedy on Q_θ: baseline policy (given by $\theta^{(0)}$) must terminate the episode!

Algorithm: Semi-gradient SARSA

Algorithm: Semi-gradient SARSA for estimating Q^π / Q^*
Input:

- a differentiable, parameter-dependent function $Q_\theta : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$
- learning rate α
- **Prediction case**: a policy π
- **Improvement case**: ϵ defining the ϵ-greedy policy update based on Q_θ

Initialize: Value function weights $\theta \in \mathbb{R}^d$ arbitrarily

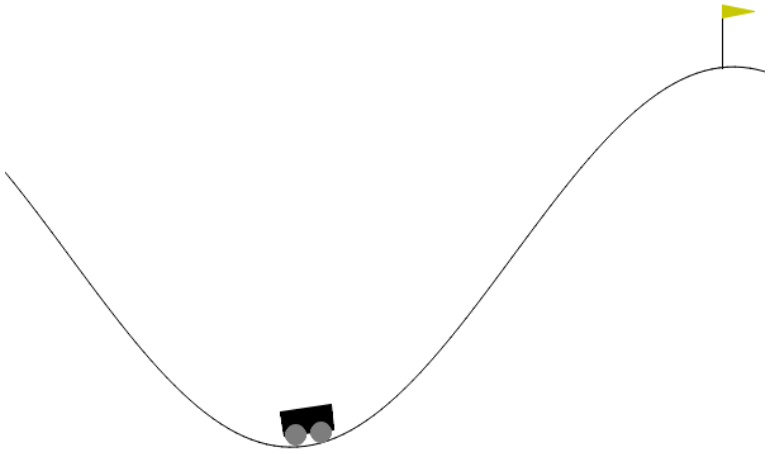
for $k = 1, 2, \dots, K$ episodes:
  Initialize s_0
  for $t = 0, 1, \dots, T-1$:
    Obtain $a_t \sim \pi(\cdot|s_t)$ (or $a_t \sim \epsilon$-greedy$(Q_\theta(s_t, \cdot))$)
    Observe r_t and s_{t+1}
    if s_{t+1} is **terminal then**
      $\theta \leftarrow \theta + \alpha[r_t - Q_\theta(s_t, a_t)]\nabla_\theta Q_\theta(s_t, a_t)$
      Go to next episode $k+1$

Choose $a_{t+1} \sim \pi(\cdot|s_{t+1})$ (or $a_{t+1} \sim \epsilon$-greedy$(Q_\theta(s_{t+1}, \cdot))$)

$$\theta \leftarrow \theta + \alpha[r_t + \gamma Q_\theta(s_{t+1}, a_{t+1}) - Q_\theta(s_t, a_t)]\nabla_\theta Q_\theta(s_t, a_t)$$

Example: Mountain car (1)

- Continous state $s \in \mathcal{S} = [x_{\text{min}}, x_{\text{max}}] \times [v_{\text{min}}, v_{\text{max}}]$ (minimal and maximal position x and velocity v).
- Single, discrete action $a \in \mathcal{A} = \{-1, 0, 1\}$.
- $r = -1$ (goal is to terminate episode as quick as possible).
- Episode terminates when car reaches the flag (or max steps).
- Simplified longitudinal car physics with state constraints.
- Position initialized randomly within valley, zero initial velocity.
- Car is underpowered and requires swing-up.

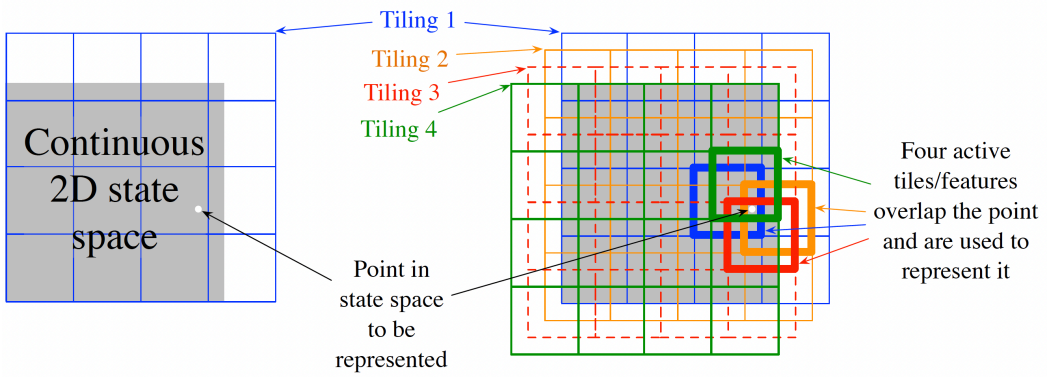


Mountain car (Sutton and Barto 1998, Gymnasium)

Approximation: Linear model with *tile coding* features $x_i(s,a)$, where

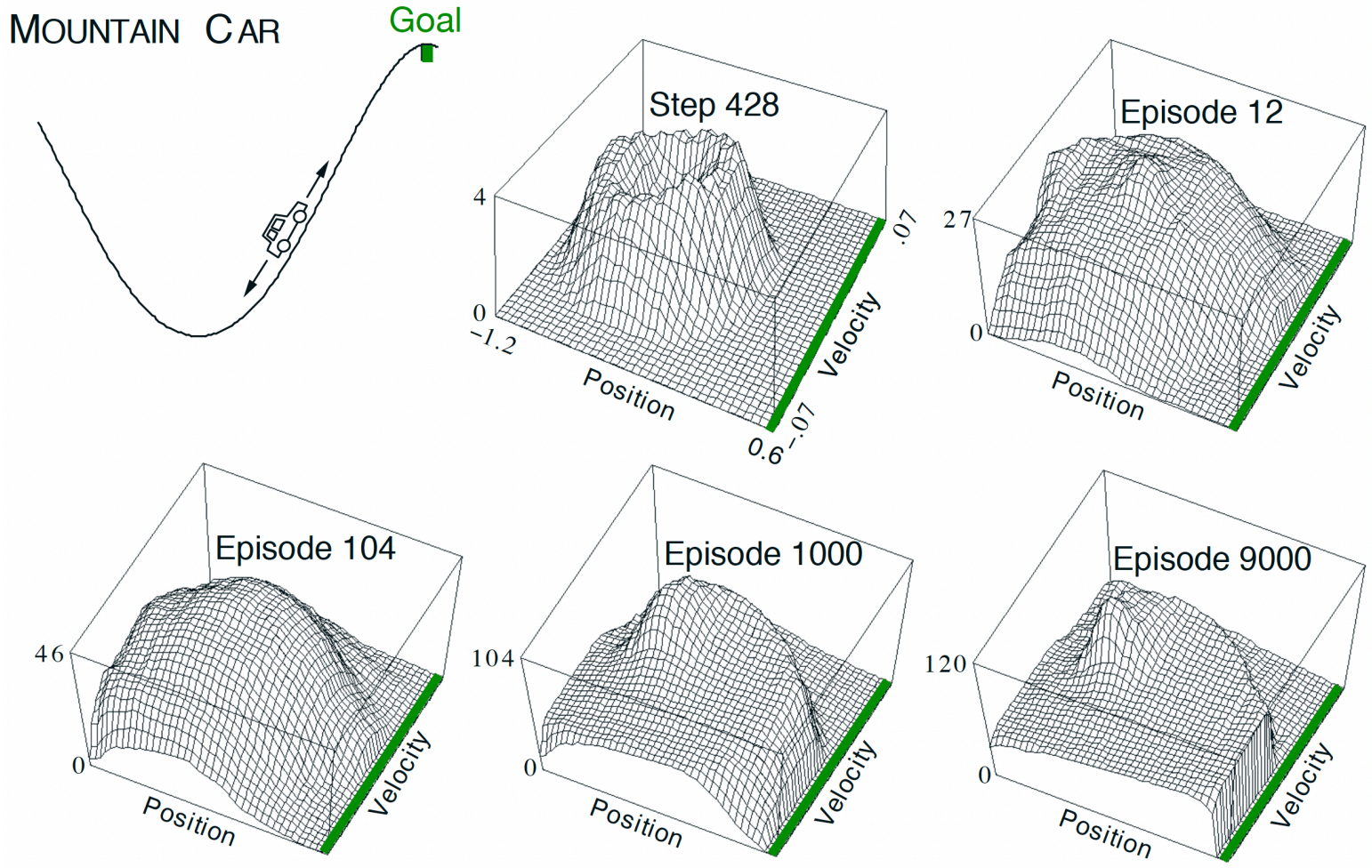
$$\psi(s,a) = \sum_{i=1}^{d} \theta_i x_i(s,a),$$

$$x_i(s,a) = \begin{cases} 1 & (s,a) \in \text{ tile \#}i \\ 0 & \text{otherwise} \end{cases}.$$



Tile coding (Sutton and Barto 1998, Figure 9.9)

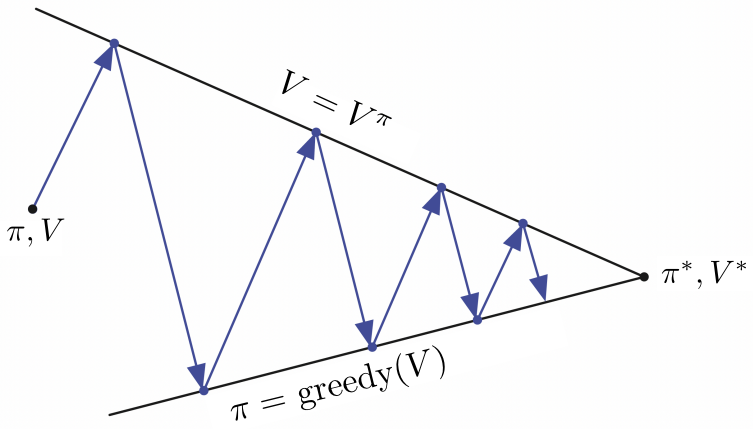
Example: Mountain car (2)



The Mountain Car task (upper left panel) and the cost-to-go function ($-\max_{a \in \mathcal{A}} Q_\theta(s, a,)$) learned during one run. (Sutton and Barto 1998, Figur 10.1)

Warning: a huge drawback of function approximation

- Recall tabular **policy improvement theorem**: guarantee to find a globally better or equally good policy in each update step.
- Parameter updates of the form (7) have an impact on $Q(s, a)$ for **all** s and a, not just the one we visited!



Generalized Policy Iteration (GPI) (Sutton and Barto 1998)

Loss of the policy improvement theorem

- When using function approximation, the policy improvement theorem does no longer apply.
- While improving the policy, we may impair it in other places *at the same time*.

Deep Q networks (DQN)

General strategy behind DQN

- Introduced in the famous DeepMind paper (Mnih et al. 2015).
- Same off-policy strategy as in the *classical/tabular* Q-learning algorithm:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a' \in \mathcal{A}} Q(s',a') - Q(s,a) \right].$$

- Now: incremental learning using semi-gradients (Eq. (7)) and TD(0) bootstrapping:

$$\theta \leftarrow \theta + \alpha \left[r + \gamma \max_{a' \in \mathcal{A}} Q_\theta(s',a') - Q_\theta(s,a) \right] \nabla_\theta Q_\theta(s,a). \tag{8}$$

But with two important distinctions

1. An **experience replay buffer** for batch learning replaces the single-sample updates.

The motivation behind this

1. **Sample efficiency**
   - *Experience replay*: efficiently use available data.
   - *Stabilize learning* by reducing the correlation between samples $\Rightarrow$ sampling from a replay buffer leads to i.i.d.-like data.
2. **Stabilization of learning**

2. A **separate set of weights** $\bar{\theta}$ for the bootstrapped Q-target ((8) is **not** a gradient).

- targets don't change in inner loop $\Rightarrow$ well-defined learning problem.

DQN working principle (1)

- Take actions u based on $Q_\theta(s, a)$ (e.g., ϵ-greedy).
- Store observed tuples (s, a, r, s') in memory buffer $\mathcal{D}$.
- Sample mini-batches $\mathcal{B}$ from $\mathcal{D}$.
- Calculate bootstrapped Q-target with a target network with weights $\bar{\theta}$:

$$Q^\pi(s, a) \approx r + \max_{a \in \mathcal{A}} Q_{\bar{\theta}}(s, a).$$

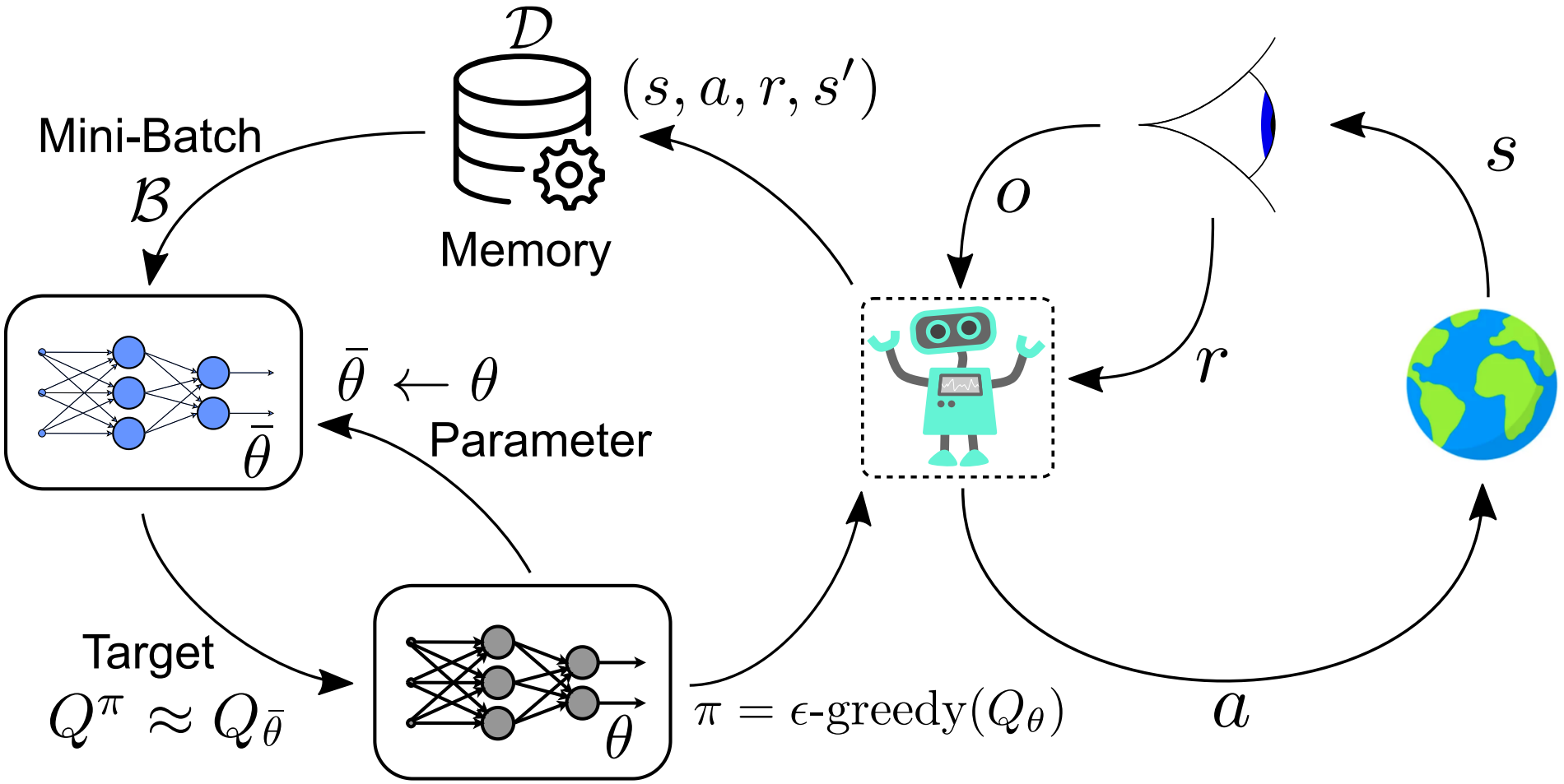
- Optimize MSE loss between above targets and the regular approximation $Q_\theta(s, a)$ using $(s_i, a_i, r_i, s_i') \in \mathcal{B}$:

$$L(\theta) = \sum_{i=1}^{|\mathcal{B}|} \Big(\underbrace{r_i + \max_{a \in \mathcal{A}} Q_{\bar{\theta}}(s_i', a)}_{= y_i \text{ (target)}} - Q_\theta(s_i, a_i) \Big)^2,$$

$$\theta \leftarrow \theta + \alpha \sum_{i=1}^{|\mathcal{B}|} [y_i - Q_\theta(s_i, a_i)] \nabla_\theta Q_\theta(s_i, a_i).$$

- Update $\bar{\theta}$ based on θ from time to time.

DQN working principle (2)



Adapted from (Abdelwanis et al. 2026)

The DQN algorithm

Algorithm: The "classic" DQN algorithm (Mnih et al. 2015)
Input:

- a differentiable, parameter-dependent function $Q_\theta : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$
- learning rate α, greedy parameter ϵ, update rate $N_\theta \in \mathbb{N}$

Initialize: Q function weights $\theta = \bar{\theta} \in \mathbb{R}^d$ arbitrarily, $\mathcal{D} = \{\}$

for $k = 1, 2, \dots, K$ episodes:
  Initialize s_0
  for $t = 0, 1, \dots, T-1$:
    Obtain $a_t \sim \epsilon$-greedy$(Q_\theta(s_t, \cdot))$ and observe r_t and s_{t+1}
    Store tuple (s_t, a_t, r_t, s_{t+1}) in $\mathcal{D}$
    Sample minibatch $\mathcal{B}$ from $\mathcal{D}$ (after initial *warm-up*)
    Compute targets for $(s_i, a_i, r_i, s_i') \in \mathcal{B}$

$$y_i = r_i + \max_{a \in \mathcal{A}} Q_{\bar{\theta}}(s_i', a) \qquad \text{(or } y_i = r_i \text{ if } s_i' \text{ is terminal)}$$

    Update θ via gradient descent on $L(\theta)$ with learning rate α
    if $t \mod N_\theta = 0$ **then** $\bar{\theta} \leftarrow \theta$

Remarks:

1. For the standard MSE loss

$$L(\theta) = \sum_{i=1}^{|\mathcal{B}|} \Big(y_i - Q_\theta(s_i, a_i)\Big)^2,$$

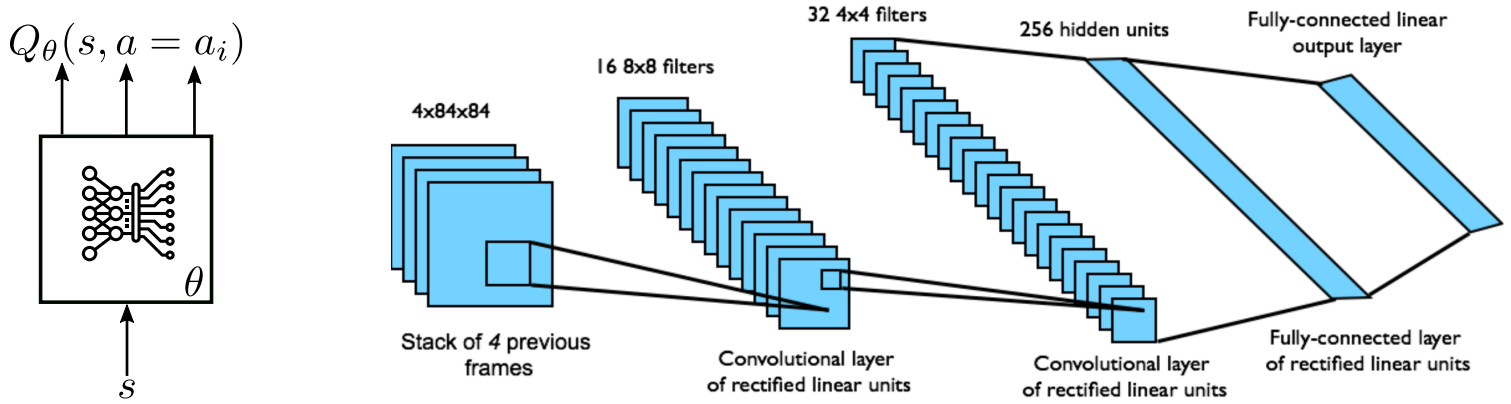
   the weight update becomes the one we have seen before:

$$\theta \leftarrow \theta + \alpha \sum_{i=1}^{|\mathcal{B}|} [y_i - Q_\theta(s_i, a_i)] \nabla_\theta Q_\theta(s_i, a_i).$$

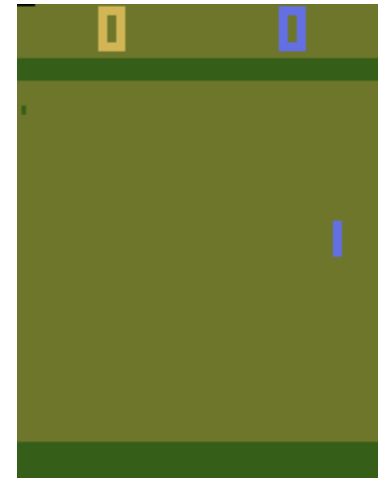
2. Apart from standard gradient descent, advanced (i.e., momentum-based) descent algorithms are possible (e.g., RMSProp, Adam, ...)

Example: Atari game Pong (1)

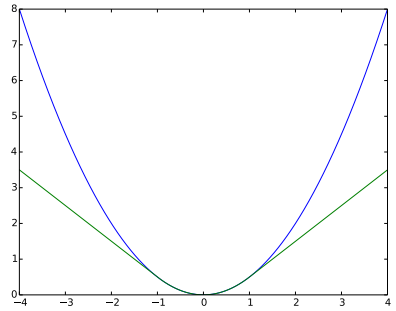
- *State* s:
  - 210×160 pixels with 3 channels each (RGB) and values $s_i \in \{0, \dots, 255\}$,
  - stacking of multiple frames,
  - some additional preprocessing.
- *Action* a: 18 actions in total, (6 buttons, some of which can be pushed at the same time).
- *Architecture* for Q_θ (Mnih et al. 2015):



- *Loss function* $L(\theta)$: *Huber loss* $L_\beta(\theta) = \begin{cases} \frac{1}{2}\theta^2, & \theta \leq \beta \\ \beta(|\theta| - \frac{1}{2}\theta), & \text{otherwise} \end{cases}$.
- *Training*: RMSProp descent algorithm.



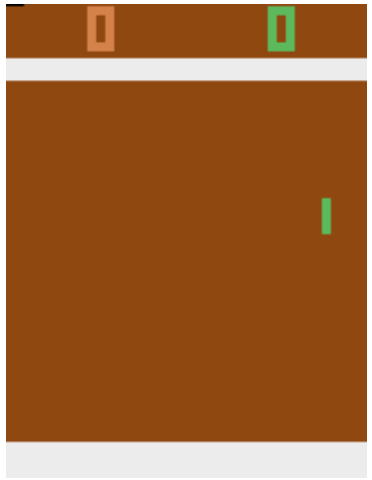
Pong setup: More details in the Farama Gymnasium description.



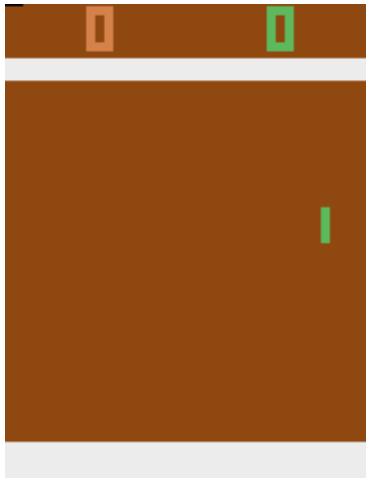
Huber loss (Wikipedia)

- *Annealing* of the exploration rate: ϵ goes from 1 to 0.1 (or 0.05) over the first 10^6 frames.

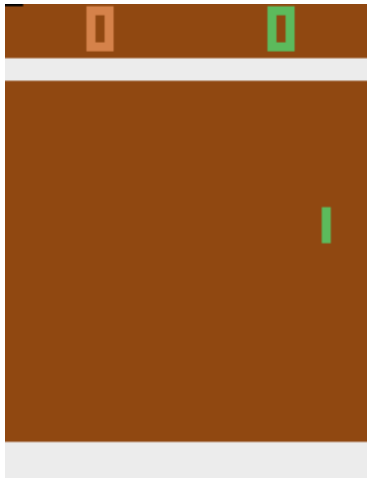
Example: Atari game Pong (2)



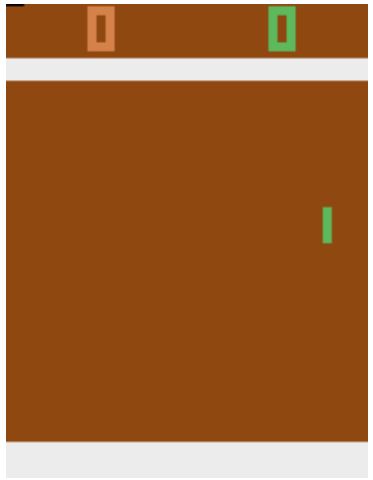
0 steps



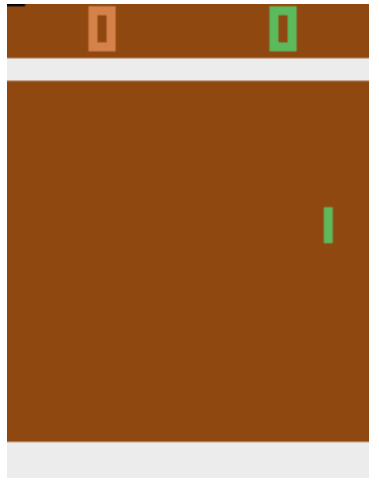
100,000 steps



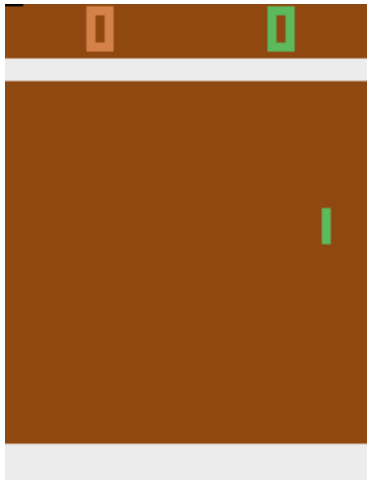
300,000 steps



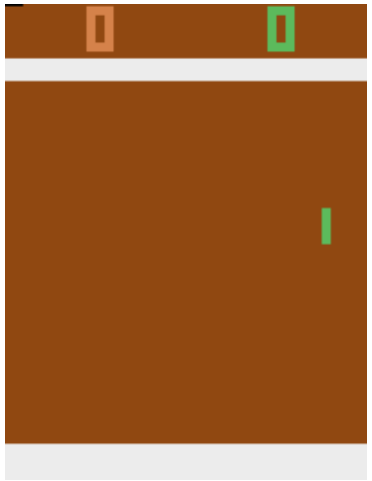
500,000 steps



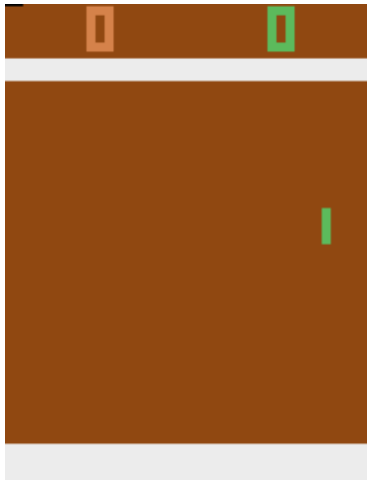
800,000 steps



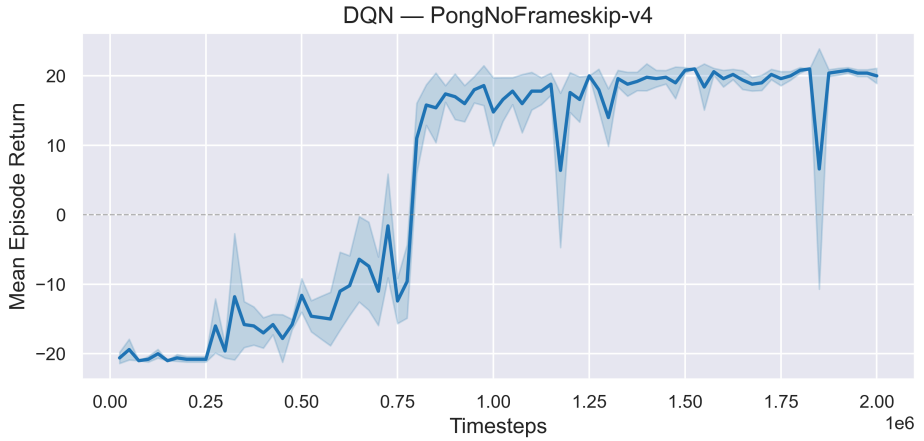
1,000,000 steps



1,500,000 steps

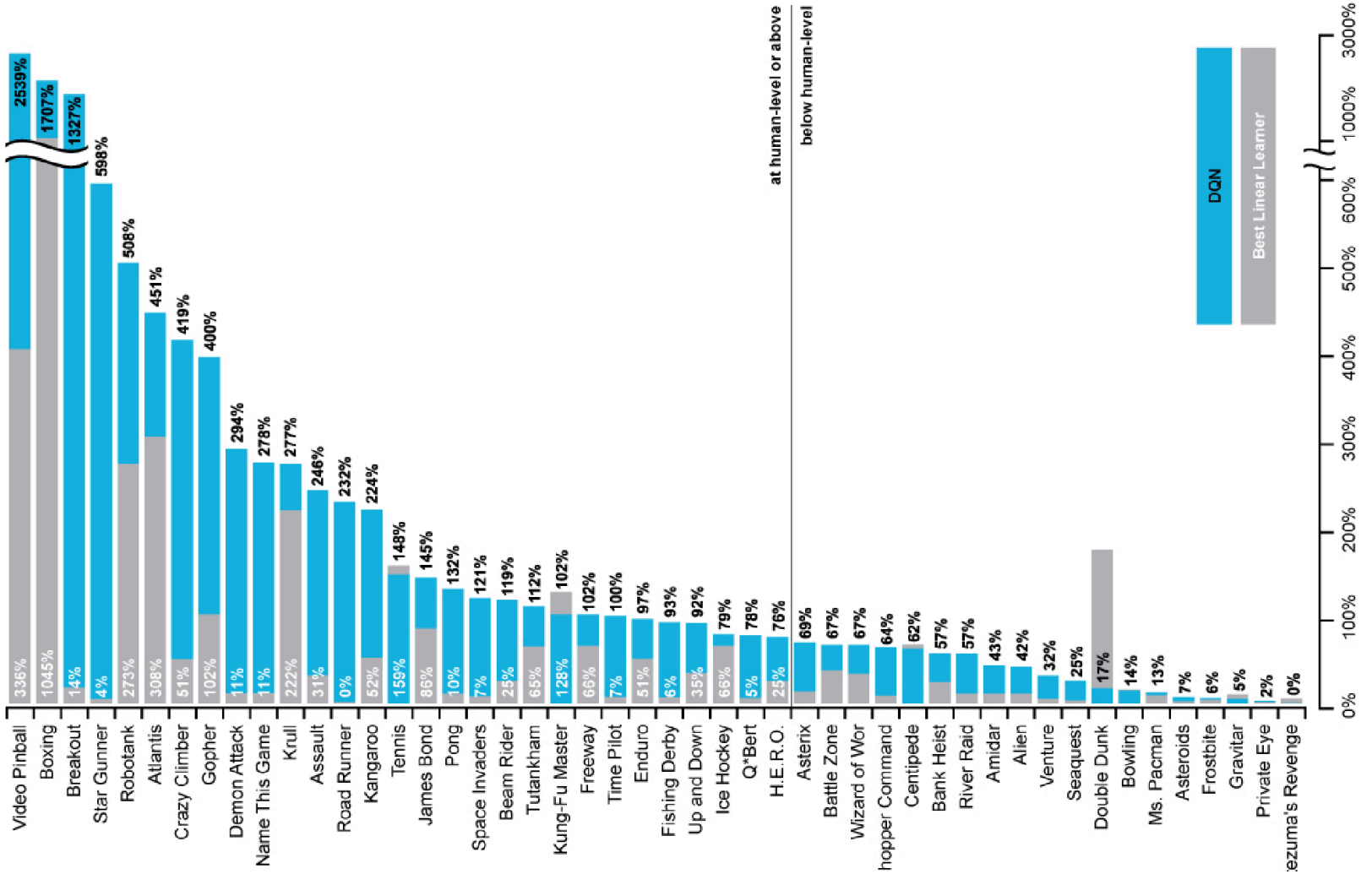


2,000,000 steps



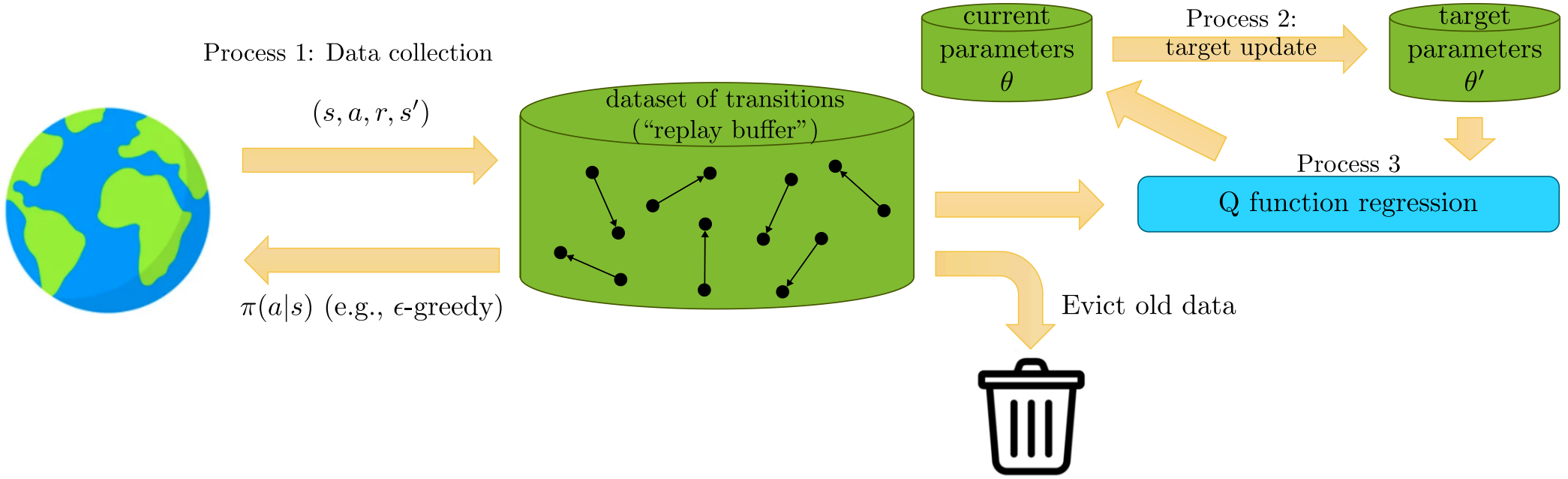
Return

Example: Atari games



Human-level control (Mnih et al. 2015)

The general view of Q-learning



Inspired by Sergey Levine's CS285 lecture.

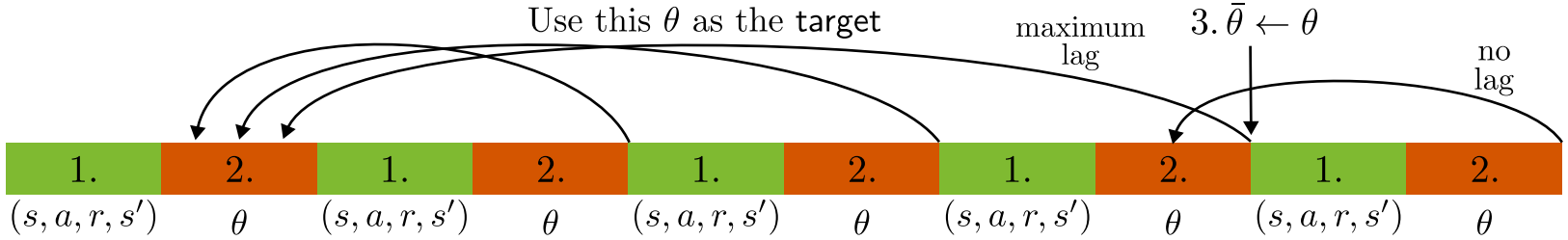
- **DQN**: Processes 1 and 3 run at the same speed, process 2 is slower.
- **Online Q-learning**: evict immediately, Process 1, 2 and 3 all run at the same speed.
  - Value or Q-learning as in the last lecture, followed directly by a policy improvement step.
  - Also similar to the value iteration algorithm from dynamic programming.

- Many variations: data collection and eviction strategies, frequencies and repetitions of the individual process steps, …

DQN extensions & modifications

Alternative target networks (Polyak averaging)

- Consider the DQN algorithm and separate it into three main components:
  1. Data collection, i.e., store tuples (s, a, r, s') in $\mathcal{D}$.
  2. Improvement using mini batches $\mathcal{B}$ and a fixed target network, e.g., $\theta \leftarrow \theta + \alpha \sum_{i=1}^{|\mathcal{B}|} [y_i - Q_\theta(s_i, a_i)] \nabla_\theta Q_\theta(s_i, a_i)$.
  3. Update target network every N_θ steps: $\bar{\theta} \leftarrow \theta$.
- Let's visualize these steps (with $N_\theta = 4$, even though it's usually much larger):



- The lag is different in different places! This doesn't need to be a problem, but it feels *uneven*.
- Alternative: **Polyak averaging**. In every step, perform the *linear interpolation*

$$\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau)\theta$$

- When initializing $\bar{\theta} = \theta$, this linear interpolation tends to work for nonlinear models (i.e., NNs) as well!

💡 Polyak averaging originates from optimization theory where in SGD, we update θ by averaging over recent iterates $\theta^{(t)}, \theta^{(t-1)}, \theta^{(t-2)}, \ldots$.

Double Q networks (Hasselt, Guez, and Silver 2016)

Remember the *maximization bias* and double Q-learning?
$\Rightarrow$ Don't use same network to choose the action and estimate the value!

Double deep Q-learning:

- Just use the current network (θ) to evaluate the action.
- Still use the target network ($\bar{\theta}$) to evaluate the value.

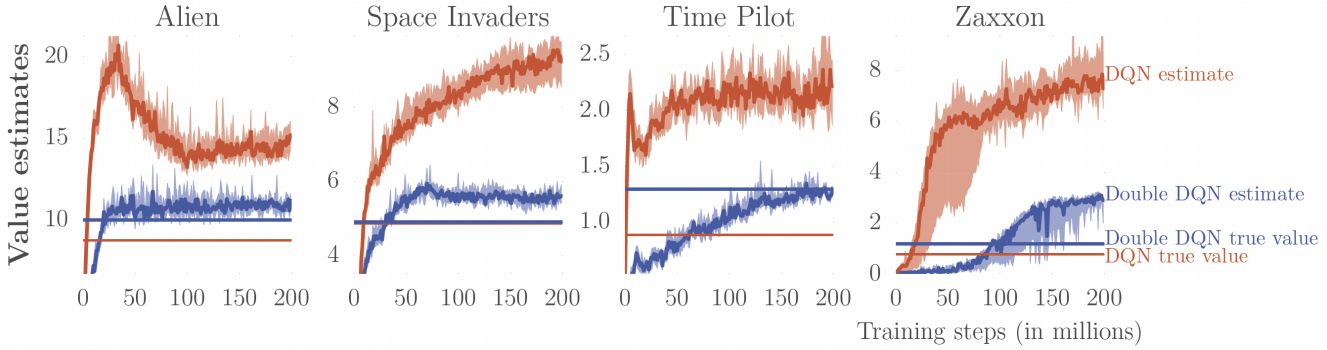
$$y = r + \gamma Q_{\bar{\theta}}(s', \arg\max_{a \in \mathcal{A}} Q_{\theta}(s', a))$$

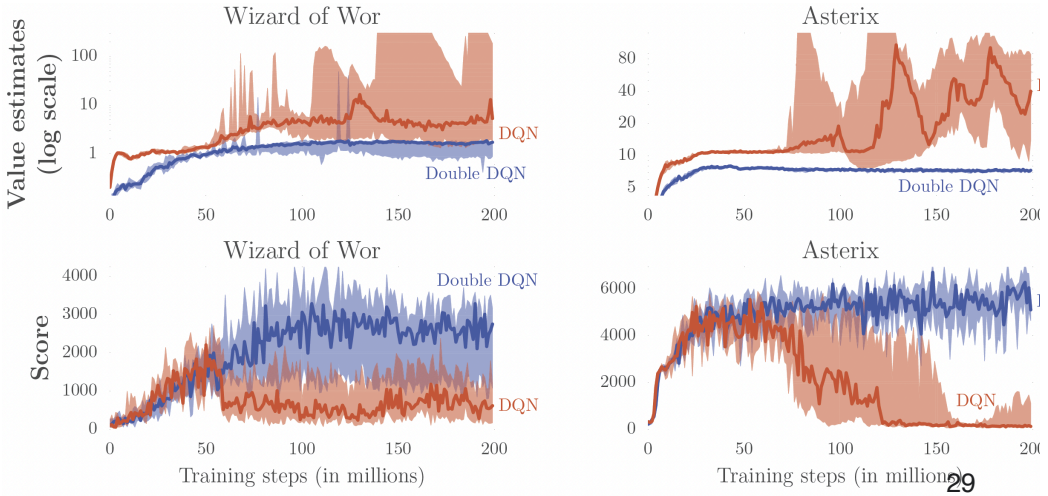
Algorithm recap: Tabular Q-Learning.

- Take a_t (ϵ-greedy on $Q_1 + Q_2$), observe (r_t, s_{t+1})
- **if rand()** > 0.5 **then**

$$y = r_t + \gamma Q_2(s_{t+1}, \arg\max_{a \in \mathcal{A}} Q_1(s_{t+1}, a))$$
$$Q_1(s_t, a_t) \leftarrow Q_1(s_t, a_t) + \alpha\,[y - Q_1(s_t, a_t)]$$

- **else**: ...

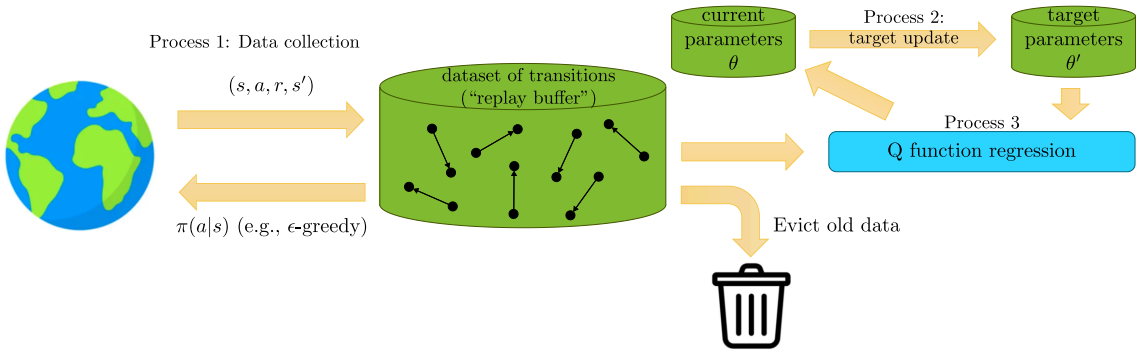




Prioritized experience replay (Schaul et al. 2016)

- Samples in the replay buffer are *not equally important*.
- From some examples, you can learn more than from others.
- What would be a good *criterion to assess the relevance* of a particular sample? ⇒ The **TD error**

$$\delta_i = r_i + \max_{a \in \mathcal{A}} Q_{\bar{\theta}}(s_i', a) - Q_\theta(s_i, a_i)$$



Inspired by Sergey Levine's CS285 lecture.

- Instead of sampling uniformly from the replay buffer, we assign an individual probability to each sample:

$$P(i) = \frac{p_i^\alpha}{\sum_{k=1}^{|\mathcal{D}|} p_k^\alpha}, \qquad p_i = |\delta_i| + \epsilon.$$

  - Here, $\alpha \geq 0$ determines the degree of prioritization ($\alpha = 0$ is the uniform case).
  - The parameter $\epsilon > 0$ ensures that all samples are drawn with non-zero probability.

- A version that's less sensitive to outliers: rank the transitions according to $|\delta_i|$:

$$p_i = \frac{1}{\mathsf{rank}(i)}.$$

- **But**: this introduces a bias (we sample from a different, uncontrollable distribution) $\Rightarrow$ Use importance sampling!

n-step returns

The concept of TD(n) can be extended to deep Q-learning in a straightforward fashion:

$$y_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \ldots + \gamma^{n-1} r_{t+n-1} + \gamma^n \max_{a \in \mathcal{A}} Q_\theta(s_{t+n}, a).$$

+ less biased target values when estimates using Q_θ are inaccurate

+ typically faster learning, especially early on

— only actually correct when learning on policy

Some remarks on continuous actions

What's the problem with continuous actions in DQN?

1. The action selection: $\pi(a|s) = \begin{cases} 1, & a = \arg\max_{a' \in \mathcal{A}} Q_\theta(s, a') \\ 0, & \text{otherwise} \end{cases}$ (or some ϵ-greedy alternative).
2. The target calculation: $y_i = r_i + \max_{a \in \mathcal{A}} Q_{\bar{\theta}}(s'_i, a)$.

$\Rightarrow$ if $\mathcal{A}$ is continuous (i.e., $a \in \mathbb{R}^m$), then the maximization becomes a **challenging optimization** probelm in itself!

What can we do?

1. Gradient based optimization (e.g., SGD) $\Rightarrow$ — This can be too slow!
2. Simple sampling:

$$\max_{a \in \mathcal{A}} Q(s, a) \approx \max\{Q(s, a_1), \dots, Q(s, a_p)\}, \qquad \text{with } \{a_1, \dots, a_p\} \sim U(\mathcal{A}).$$

   $\Rightarrow$ + Very efficient. + Easily parallelized. — Not very accurate.

3. Other stochastic optimization techniqes $\Rightarrow$ — Usually also quite slow, or scale poorly.

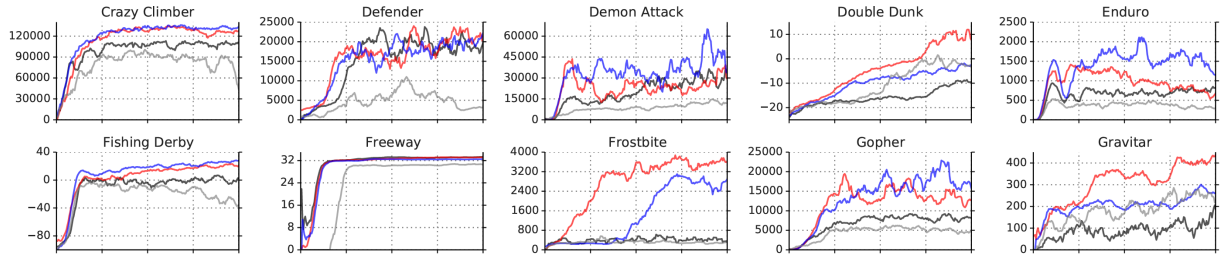
4. Train second network $\pi_\phi : \mathcal{S} \to \mathcal{A}$ whose output maximizes $Q_\theta(s, a)$:

$$Q(s, \pi_\phi(s)) \approx \max_{a \in \mathcal{A}} Q_\theta(s, a) \qquad \Rightarrow \quad \text{Actor-critic methods (in two lectures)!}$$

Some practical tips

Following Sergey Levine's CS285 lecture:

- Q learning takes some care to stabilize.
  - Test on easy, reliable tasks first, make sure your implementation is correct.



(*Schaul et al. 2016*)

- Large replay buffers help improve stability.
- Looks more like fitted Q iteration.
- It takes time, be patient; performance might be no better than random for a while.
- Start with high exploration (ϵ) and gradually reduce.
- Bellman error gradients can be big; clip gradients or use Huber loss.
- Double Q learning helps a lot in practice: it's simple and has no downsides.
- n-step returns also help a lot, but have some downsides.
- Schedule exploration (high to low) and learning rates (high to low). The Adam optimizer can help too.
- Run multiple random seeds, it's very inconsistent between runs.

Least squares policy iteration

Linear action-value-function estimation

- Recall the LSTD algorithm from last lecture: batch least squares estimation $V^\pi(s)$ with linear models:

$$y = \begin{bmatrix} r_0 \\ \vdots \\ r_{T-1} \end{bmatrix}, \quad \Psi = \begin{bmatrix} \psi_0^\top - \gamma\psi_1^\top \\ \vdots \\ \psi_{T-1}^\top - \gamma\psi_T^\top \end{bmatrix}, \quad L(\theta) = \frac{1}{N}\|y - \Psi\theta\|_F^2, \quad \theta^* = (\Psi^\top\Psi)^{-1}\Psi^\top y = \Psi^\dagger y, \quad V_{\theta^*}(s) = \theta^{*\top}\psi(s).$$

- Let's do the same for Q values $\Rightarrow$ **LS-SARSA** / **LSTD$\mathbf{Q}$**

$$Q_\theta(s,a) = \theta^\top\psi(s,a) = \sum_{i=1}^d \theta_i\psi_i(s,a) \qquad \text{(Linear model)}$$

$$Q^\pi(s,a) \approx r + \gamma Q_\theta(s',a') \qquad \text{(TD(0) bootstrapping)}$$

- Loss function:

$$L(\theta) = \sum_{t=0}^{T}\left(r_t - \left[\psi_t^\top - \gamma\psi_{t+1}^\top\right]\theta\right)^2 = \frac{1}{N}\|y - \Psi\theta\|_F^2, \qquad \text{where } \psi_t = \psi(s_t, a_t).$$

- Same as before, only that the features depend on both states and actions: $\psi(s,a)$.

On-policy and off-policy LS-SARSA

- Same data collection procedure as before: batch mode $\Rightarrow \Psi$ and y.
- Regarding the data in Ψ, we can distinguish two cases:
  - the same policy π (*on-policy learning*),
  - an arbitrary policy $a' \sim \pi'(\cdot|s')$ (*off-policy learning*).
- If we apply off-policy LS-SARSA then
  - we retrieve the flexibility to collect training samples arbitrarily,
  - at the cost of an estimation bias based on the sampling distribution.
- Possible modifications:
  - To prevent numeric instability regularization is possible, e.g., ridge regression $\theta^* = (\Psi^\top \Psi + \epsilon I)^{-1} \Psi^\top y$
  - Recursive implementation for online usage is straightforward.

Least squares policy iteration (LSPI)

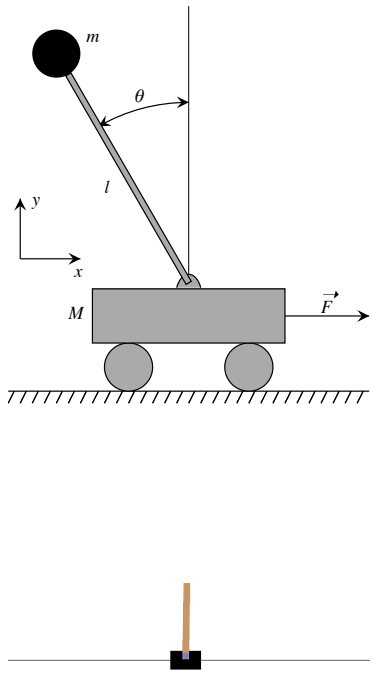
General idea:

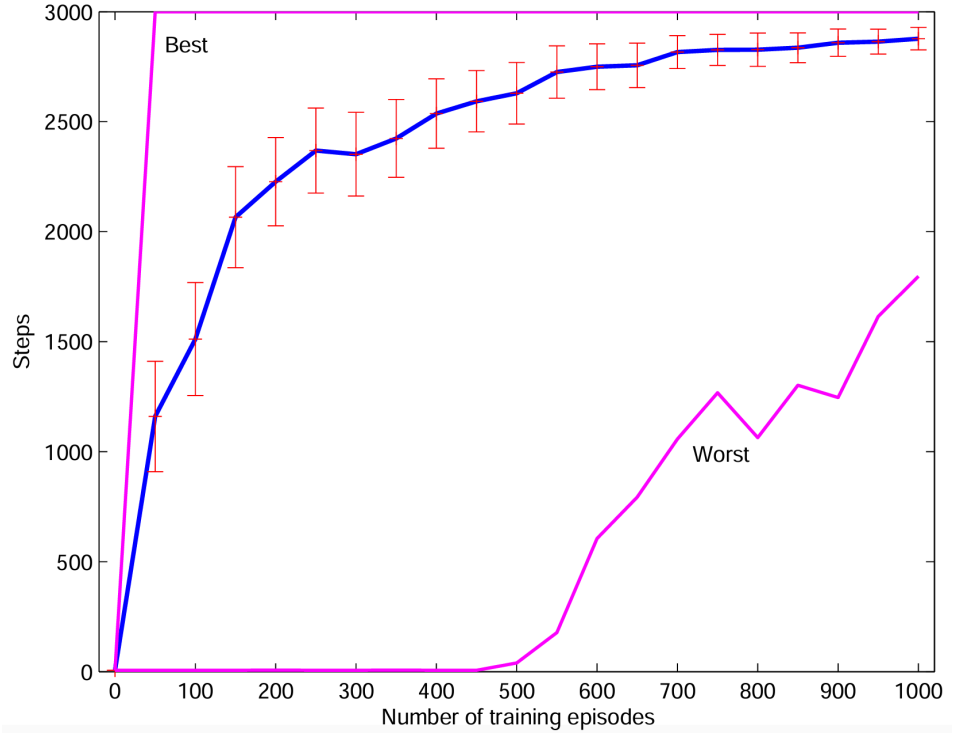
- Apply general policy improvement (GPI) based on dataset $\mathcal{D}$,
- Policy evaluation by off-policy LS-SARSA,
- Policy improvement by greedy choices on predicted action values.

Some remarks:

- LSPI is an offline and off-policy control approach.
- Exploration is required by feeding suitable sampling distributions to $\mathcal{D}$:
  - ϵ-greedy choices based on Q_θ.
  - Completely random samples are conceivable as well.

Example: Pendulum on a cart (Abdelwanis et al. 2026)





Balancing steps before episode termination with a clipping of maximum 3000 steps (Lagoudakis and Parr 2003)

Summary / what you have learned

Summary / what you have learned

- The procedures from the approximate prediction can be transferred to value-based control in a straightforward way.
- Downside: the policy improvement theorem no longer applies in the approximate RL case.
  - Control algorithms may diverge.
  - Performance trade-offs between different parts of state-action space could emerge.
- Off-policy batch learning approaches allow for efficient data usage.
  - LSPI uses LS-SARSA on linear function approximation.
  - DQN extends Q-learning on non-linear approximation with additional tweaks (experience replay, target networks,...).
  - However, a prediction bias results (off-policy sampling distribution).

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